Bees 2: Varroa mites and deformed wing virus

Simon Frost

Table of contents

Mites are a three-stage net 1
Bees and mites meet through pools 2
The virus is a stratum 4
Treatment, and its timing 7
Where the numbers come from 8
References 8

```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Plots
```

The previous vignette built a honey bee colony. This one adds the mite *Varroa destructor* as a second species, and then deformed wing virus (DWV) as a second *stratum* of both species.

Mites are a three-stage net

A mite is on an adult bee (`phoretic`) or in a brood cell, where it stays until the bee emerges and then leaves with its daughters: 12 days in a worker `cell`, 15 in a `drone_cell`. Drone cells are where mites do best. A mite there leaves 2.91 daughters, against 1.01 in a worker cell, and fewer mites are lost when the bee emerges.

```
q = VARROA_PARAMETERS
mite_life = ValuedProjectionNet([:phoretic, :cell, :drone_cell],
    :waiting         => [(:phoretic => :phoretic) => 1.0],
    :invading        => [(:phoretic => :cell) => 1.0],
    :leaving         => [(:cell => :phoretic) => 1 - q.emergence_loss],
    :daughters       => [(:cell => :phoretic) => q.offspring * (1 -
     ↪  q.emergence_loss)],
    :invading_drone  => [(:phoretic => :drone_cell) => 1.0],
    :leaving_drone   => [(:drone_cell => :phoretic) => 1 -
     ↪  q.drone_emergence_loss],
```

```
    :daughters_drone => [(:drone_cell => :phoretic) => q.drone_offspring *
     ↪  (1 - q.drone_emergence_loss)])
length(stage_names(expand_stages(mite_life, :cell => 12, :drone_cell =>
 ↪  15)))
```

```
28
```

`varroa()` is this net with responses. The important ones are on the two `invading` transitions. The rate at which phoretic mites enter cells is 0.56 × worker cells about to be capped / grams of adult bees, plus 6.49 × drone cells / grams of adult bees (Boot et al. 1995; Calis et al. 1999): a drone cell attracts almost twelve times as strongly as a worker cell. A cell takes at most four mites, and mites sharing a cell leave fewer daughters each (factors of 1, 0.91, 0.86 and 0.60 for one to four mites in a worker cell; Martin 1998). Mites on bees die at 0.006 a day while the colony has brood and at 0.002 a day while it has none.

Bees and mites meet through pools

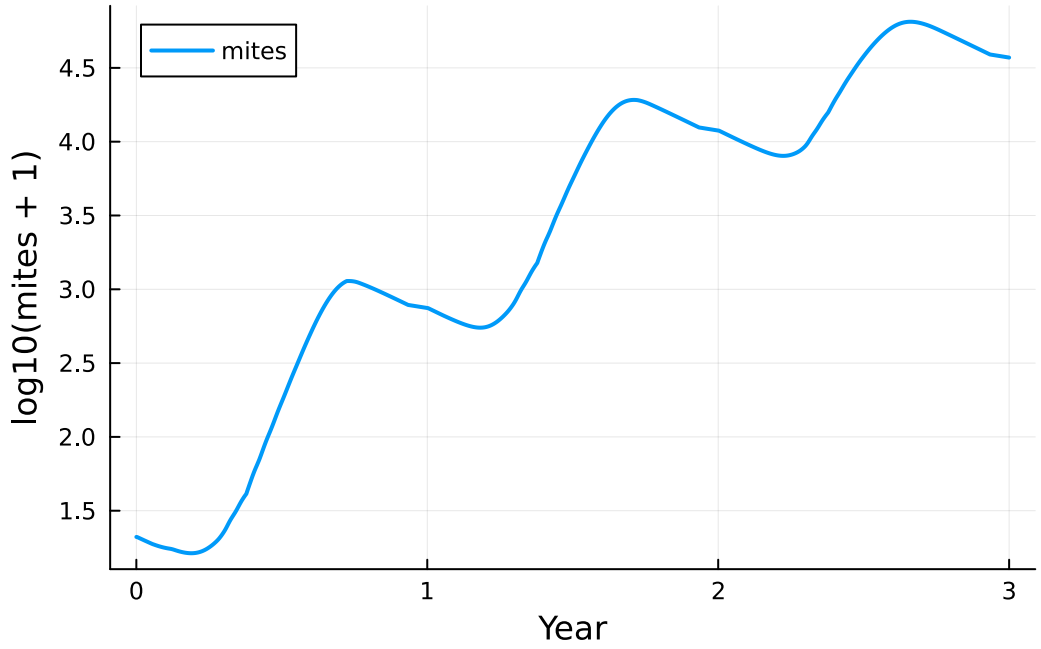
The mite never refers to the bee, or the bee to the mite. The bee reports a few numbers as pools, and the mite reads them.

```
c = community(honeybee(), varroa())
pool_names(c)
```

```
7-element Vector{Symbol}:
 :adult_bees
 :brood
 :capping
 :emerging
 :drone_capping
 :drone_emerging
 :phoretic
```

`capping` and `drone_capping` are the worker and drone larvae about to be capped, `emerging` and `drone_emerging` the pupae about to emerge, `adult_bees` the adult workers, `brood` all the brood, and `phoretic` the mites on adults.

```
sol = project(c, honeybee_state(c; mites = 20), 1:(365 * 3))
years = sol.t ./ 365
mites = abundance(sol, :varroa)
adults(s) = stage_total(s, :honeybee, [:hive, :forager])
plot(years, log10.(mites .+ 1); label = "mites", xlabel = "Year", ylabel =
 ↪  "log10(mites + 1)", lw = 2)
```




```
(growth_per_day = round(log(mites[211] / mites[151]) / 60; digits = 4),
 ↪  fold_in_first_year = round(mites[366] / 20; digits = 1))
```

```
(growth_per_day = 0.0311, fold_in_first_year = 37.3)
```

Drone brood is a small part of the brood, about one cell in forty, but it carries the mite population. The same colony without drones:

```
w = community(honeybee(drones = false), varroa(drones = false))
mites_w = abundance(project(w, honeybee_state(w; mites = 20), 1:(365 * 3)),
 ↪  :varroa)
(growth_per_day = round(log(mites_w[211] / mites_w[151]) / 60; digits = 4),
 ↪  fold_in_first_year = round(mites_w[366] / 20; digits = 1))
```

```
(growth_per_day = 0.0181, fold_in_first_year = 8.4)
```

```
share = stage_total(sol, :varroa, :drone_cell)[180] / mites[180]
round(share; digits = 2)       # of all mites, in drone cells on day 180
```

```
0.19
```

Nothing here was tuned. Martin (1998) reports a 12-fold increase a year for his model, which includes drone brood; this one grows three times as fast as that with drones and somewhat more slowly without. It behaves like BEEHAVE, whose rules it follows: BEEHAVE runs started with 20 mites are reported to peak near 24,000 in the fourth summer, and removing drone brood in BEEHAVE cuts mite numbers by 90% in a year (Schödl et al. 2022). Here the mites get there

about a year sooner. Schödl et al. note that bee experts and field studies put the effect of removing drone brood at 50 to 67%, so these rules probably give drone brood too much weight.

In this model mites alone do not harm the bees, following BEEHAVE, in which all damage comes through viruses:

```
alone = project(honeybee(), honeybee_state(honeybee()), 1:(365 * 3))
maximum(abs.(adults(sol) .- adults(alone)))
```

```
0.0
```

The virus is a stratum

DWV divides bees and mites by infection. That is a stratification of each life cycle, not a new model: every stage and transition is replicated, and a *coupling* says which transitions can cross from one stratum to another. Bees have three strata: `healthy`, `carrier` (infected as an adult) and `infected` (infected as a pupa).

```
healthy_only = honeybee()
with_virus = honeybee(virus = true)
length(stage_names(healthy_only, :honeybee)), length(stage_names(with_virus,
 ↪  :honeybee))
```

```
(49, 143)
```

```
filter(t -> occursin("_to_", String(t)),
 ↪  transition_names(system(with_virus)[:honeybee]))
```

```
5-element Vector{Symbol}:
 :capping_healthy_to_infected
 :drone_larva_ageing_healthy_to_infected
 :foraging_healthy_to_carrier
 :hive_life_healthy_to_carrier
 :recruiting_healthy_to_carrier
```

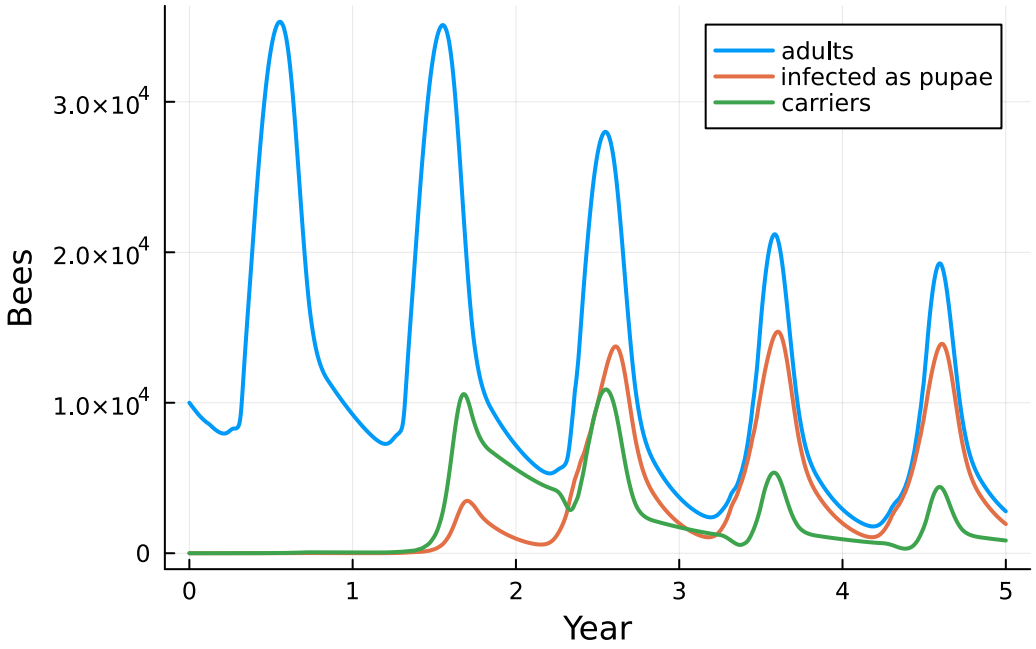
A larva becomes `infected` when its cell is closed, if a mite carrying the virus has entered it. A fifth of infected pupae die, and the adults they become die three times as fast in the hive. An in-hive bee becomes a `carrier` when a mite with the virus takes it as a host. Carriers live as long as healthy bees (for this virus; BEEHAVE), but a mite feeding on one picks the virus up. (Adults can also become carriers by mouth, used in the last vignette.)

Mites have two strata, `clean` and `viral`. A clean mite becomes viral in a cell whose pupa is infected, or by taking a bee with the virus as its host. A mite takes a new host when it leaves a cell, or when its host dies or becomes a forager, and the chance that the new host has the virus is the fraction of in-hive bees that do. A viral mite stays viral; its daughters are viral if the pupa in their cell was infected and lived.

```
filter(t -> occursin("_to_", String(t)),
 ↪  transition_names(system(varroa(virus = true))[:varroa]))
```

```
9-element Vector{Symbol}:
 :daughters_clean_to_viral
 :daughters_drone_clean_to_viral
 :daughters_drone_viral_to_clean
 :daughters_viral_to_clean
 :invading_clean_to_viral
 :invading_drone_clean_to_viral
 :leaving_clean_to_viral
 :leaving_drone_clean_to_viral
 :waiting_clean_to_viral
```

```
function colony(; treatment = day -> 0.0, years = 5)
    c = community(honeybee(virus = true), varroa(virus = true, treatment =
 ↪  treatment))
    project(c, honeybee_state(c; mites = 20, viral = 0.1), 1:(365 * years))
end
untreated = colony()
infected(s) = stage_total(s, :honeybee, [:hive_infected, :forager_infected])
carriers(s) = stage_total(s, :honeybee, [:hive_carrier, :forager_carrier])
plot(untreated.t ./ 365, [adults(untreated) infected(untreated)
 ↪  carriers(untreated)];
     label = ["adults" "infected as pupae" "carriers"], xlabel = "Year",
 ↪  ylabel = "Bees", lw = 2)
```



For a year little happens. Carriers come first: by the second winter most bees carry the virus without harm. From the second autumn, when mites are numerous and brood is scarce, a growing share of the bees that will form the winter cluster are infected as pupae. They die early, and the colony goes into each following winter with a few thousand bees and more mites than bees.

```
winters = 365 .* (1:5) .+ 1
[(year = y, adults = round(Int, adults(untreated)[w]), mites_per_100_bees =
 ↪  round(100 * stage_total(untreated, :varroa, :phoretic)[w] /
 ↪  adults(untreated)[w]; digits = 1),
  infected = round(infected(untreated)[w] / adults(untreated)[w]; digits =
 ↪  2),
  carriers = round(carriers(untreated)[w] / adults(untreated)[w]; digits =
 ↪  2)) for (y, w) in enumerate(winters)]
```

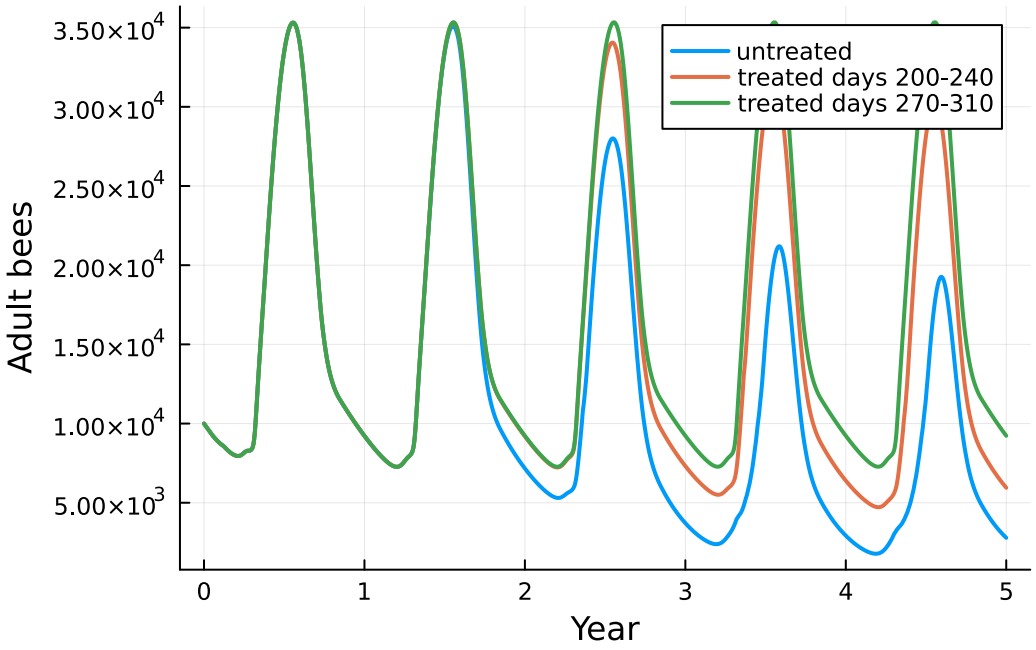
```
5-element Vector{@NamedTuple{year::Int64, adults::Int64,
 ↪  mites_per_100_bees::Float64, infected::Float64, carriers::Float64}}:
 (year = 1, adults = 9218, mites_per_100_bees = 7.7, infected = 0.0,
  ↪  carriers = 0.01)
 (year = 2, adults = 7165, mites_per_100_bees = 128.8, infected = 0.14,
  ↪  carriers = 0.78)
 (year = 3, adults = 3714, mites_per_100_bees = 561.8, infected = 0.53,
  ↪  carriers = 0.46)
 (year = 4, adults = 2923, mites_per_100_bees = 789.9, infected = 0.68,
  ↪  carriers = 0.32)
 (year = 5, adults = 2785, mites_per_100_bees = 789.7, infected = 0.7,
  ↪  carriers = 0.3)
```

A colony of fewer than about 4,000 bees is usually taken not to survive the winter (BEEHAVE; Kielmanowicz et al. 2015), so this colony is effectively lost in its third winter. Field studies put the autumn load of colonies that died at about 15 mites per 100 bees against 3 in survivors (Genersch et al. 2010).

Treatment, and its timing

A treatment is an extra mortality of phoretic mites over a window of days. Two windows of 40 days, each year:

```
late = colony(treatment = day -> 270 <= day < 310 ? 0.115 : 0.0)
early = colony(treatment = day -> 200 <= day < 240 ? 0.115 : 0.0)
plot(untreated.t ./ 365, [adults(untreated) adults(early) adults(late)];
     label = ["untreated" "treated days 200–240" "treated days 270–310"],
 ↪  xlabel = "Year", ylabel = "Adult bees", lw = 2)
```




```
[(treatment = name, winter_5_adults = round(Int, adults(s)[winters[5]]),
 ↪  mites = round(Int, abundance(s, :varroa)[winters[5]]))
 for (name, s) in (("none", untreated), ("days 200–240", early), ("days
 ↪  270–310", late))]
```

```
3-element Vector{@NamedTuple{treatment::String, winter_5_adults::Int64,
 ↪  mites::Int64}}:
 (treatment = "none", winter_5_adults = 2785, mites = 21989)
```

```
 (treatment = "days 200–240", winter_5_adults = 5946, mites = 4116)
 (treatment = "days 270–310", winter_5_adults = 9229, mites = 0)
```

Treating in October, when there is almost no brood for the mites to hide in, removes nearly all of them and keeps the colony healthy. The same treatment in late July and August kills only the mites on adults at the time; those in cells survive it, the population recovers, and the colony dwindles, more slowly than without treatment. (Field advice is to treat before the winter bees are reared; in this model the absence of brood matters more, which may reflect its lack of mites arriving from other colonies.)

Where the numbers come from

The mite and virus parameters are those of Martin (1998, 2001), checked against the parameter list and program code of BEEHAVE (Becher et al. 2014), which cite them:

- daughters per mite, 1.01 in a worker cell and 2.91 in a drone cell;
- mites lost at emergence, 30% and 20%;
- the factors for mites sharing a cell, and the limit of four mites to a cell;
- mite mortality of 0.006 a day with brood and 0.002 without;
- transmission from mite to pupa of 0.89, death of a fifth of infected pupae, and adult mortality of 0.012 a day after infection as a pupa;
- a treatment killing 11.5% of phoretic mites a day for 40 days.

The drone parameters of the colony (one drone egg to 25 worker eggs from day 115 to 240, development times, survival) are BEEHAVE's too. All are listed in `VARROA_PARAMETERS` and `HONEYBEE_PARAMETERS`. Martin's papers themselves were not read. Of Schödl et al. (2022) only the abstract was read, and the figure of 24,000 mites is from a beekeeping article describing BEEHAVE runs, not from a paper.

The rules follow BEEHAVE's, in expectation: BEEHAVE draws whole mites and whole bees at random, and this model moves the expected numbers, taking mites to be spread over cells at random. Two simplifications remain. Mites that take a host are counted without allowing for crowding in the cell they left. And when brood dies in its cell the mites are held there until the bee would have emerged, where BEEHAVE releases them at once.

References

Martin, S. J. (1998). *Ecological Modelling* 109, 267-281; (2001) *Journal of Applied Ecology* 38, 1082-1093. Becher, M. A. et al. (2014). *Journal of Applied Ecology* 51, 470-482. Genersch, E. et al. (2010). *Apidologie* 41, 332-352. Schödl, I. et al. (2022). *Ecology and Evolution* 12, e9456.